



# Probing Functional Correctness in Diffusion Language Models

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[github.com/guan404ming/dllm-probing](https://github.com/guan404ming/dllm-probing)



## Motivation

**Diffusion language models** like LLaDA and Dream generate text by *iterative denoising* of all tokens in parallel, not left-to-right. AR probing literature is mature; DLM internals are unexplored, and bidirectional multi-step denoising creates a *fundamentally different* representational landscape.

**Question.** *When* during denoising, and *where* in the layer stack, do hidden states encode whether the output will be functionally correct?

**Why it matters.** The same probe yields a practical lever, *selective generation*, that skips denoising on instances predicted to fail – saving **36–98%** of wasted compute and beating softmax heuristics by 5–20 pts.

## Method & Setup

**Models.** LLaDA-8B-Instruct (33 layers, block-masked) and Dream-7B-Instruct (29 layers, global-linear).

**Tasks** (structural  $\rightarrow$  reasoning): JSON Schema (272), GSM8K (1,319), MBPP (257), ARC-Challenge (1,172).

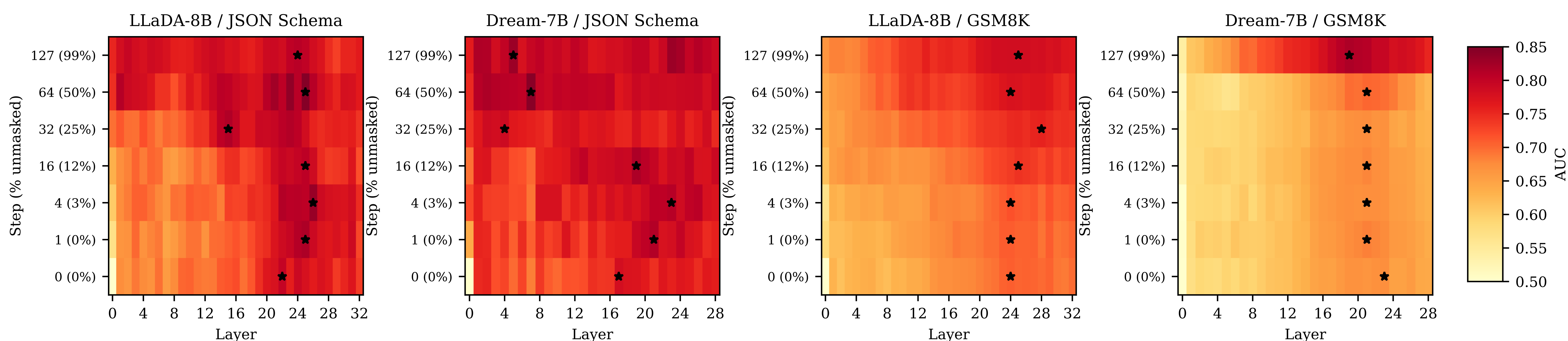
**Probing pipeline** on a frozen DLM:

- Run 128-step denoising; extract hidden states at 7 checkpoints  $t \in \{0, 1, 4, 16, 32, 64, 127\}$ .
- Partition generation region into 4 regions, mean-pool  $\rightarrow$  feature  $(L, 4, d)$  per step.
- Per (step, layer, region): PCA(64)  $\rightarrow$  *logistic regression*; AUC under 5-fold stratified CV.

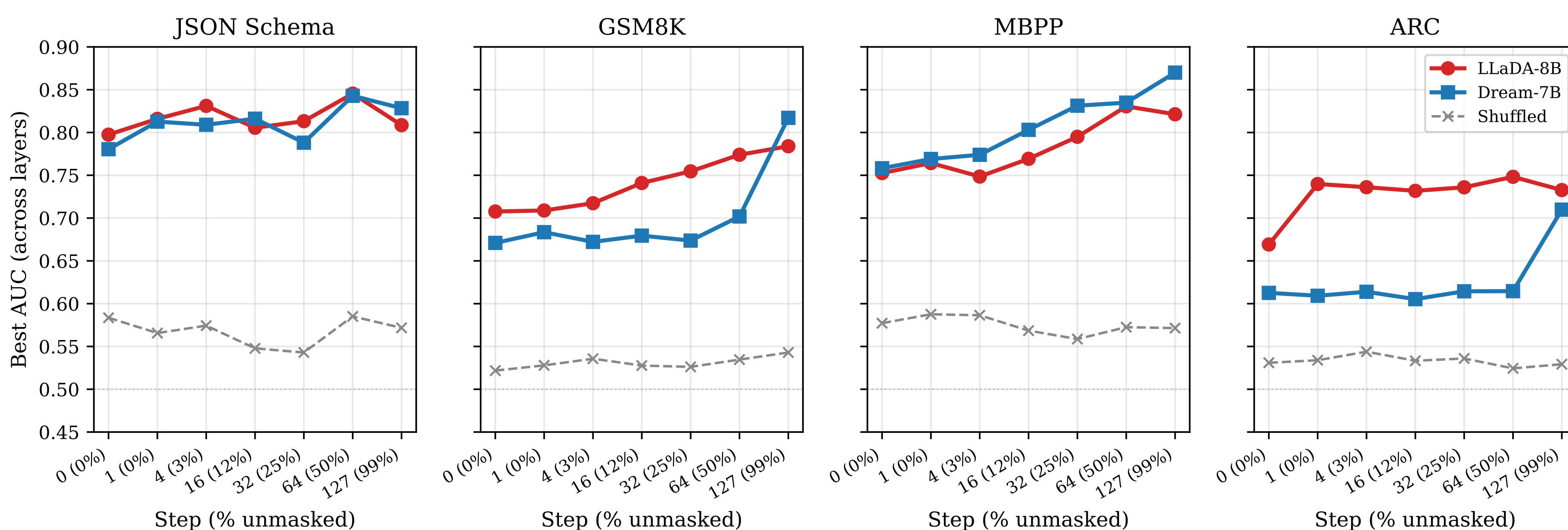
## Key Findings

- DLMs **uniquely accumulate** correctness signal during denoising ( $\Delta$ AUC 0.08–0.11 on reasoning) – a capacity *absent* in single-pass AR decoding.
- Step-0 already carries strong above-chance signal (AUC 0.61–0.80), largely reflecting **prompt difficulty**, not diffusion-specific computation.
- Emergence is **task-dependent**: structural tasks flat from step 0 (AUC  $\approx$  0.80); reasoning tasks build gradually across denoising.
- Divergent layer dynamics**: LLaDA concentrates in upper layers (L22–28); Dream *migrates downward* on simple tasks (L17 $\rightarrow$ L4), tracking pretraining origin.

## AUC heatmaps: layer $\times$ denoising step



## Best AUC across layers at each step



## Selective Generation ( $\tau=0.80$ )

|                 | JSON        | GSM8K | MBPP        | ARC         |
|-----------------|-------------|-------|-------------|-------------|
| <i>LLaDA-8B</i> |             |       |             |             |
| Compute saved   | <b>98.4</b> | 61.3  | <b>95.5</b> | <b>82.9</b> |
| Clf. accuracy   | 73.5        | 73.5  | 66.5        | 79.6        |
| Majority acc.   | 51.5        | 66.3  | 59.1        | 78.2        |
| <i>Dream-7B</i> |             |       |             |             |
| Compute saved   | <b>96.7</b> | 36.0  | <b>97.8</b> | 61.4        |
| Clf. accuracy   | 72.4        | 73.5  | 70.0        | 75.9        |
| Majority acc.   | 54.0        | 61.5  | 55.6        | 74.5        |

Dashed grey is a shuffled-label control (peaks  $\leq$  0.58). The four tasks form a clean *structural* $\rightarrow$ *reasoning* ladder: JSON is flat ( $\approx$  0.80 from step 0); GSM8K, MBPP, ARC climb 0.08–0.11 AUC across denoising. LLaDA (red) leads on most tasks; Dream (blue) catches up at full denoising.

**Sv**: % of denoising steps skipped when probe predicts failure. **Ac**: classification accuracy of probe-as-gate. JSON/MBPP: skip  $>$  95% of compute while clearing the majority baseline by 7–22 pts. All values in %.

## Controls

- Shuffled labels**: AUC  $\leq$  0.58 everywhere; *prompt-length-matched*: avg. drop  $\leq$  0.02.
- Length-matched probe**: AUC drops only 0.00–0.09 after matching output length; MBPP shows *no* drop.
- AR baseline** on Qwen-2.5-7B last-prompt token matches DLM step-0 on most tasks, confirming the step-0 signal reflects **prompt difficulty**, not diffusion-specific computation.

## Conclusion

- First probing study* of DLM hidden states for functional correctness across the full (layer, step, region) grid for 2 models  $\times$  4 tasks.
- Iterative denoising provides DLMs with  $\Delta$ AUC 0.08–0.11 on reasoning that single-pass AR decoding cannot reach.
- Probe confidence enables *offline selective generation* that saves **36–98%** compute and beats softmax heuristics by 5–20 pts.

## Limitations & Future Work

- Linear probes with PCA features; *nonlinear* or multi-layer probes may capture richer structure.
- Selective generation evaluated *offline*; live deployment requires probe-cost accounting and threshold calibration.
- Layer-dynamics is correlational; a controlled AR-base fine-tuning ablation, *continuous diffusion LMs*, and larger scales remain open.